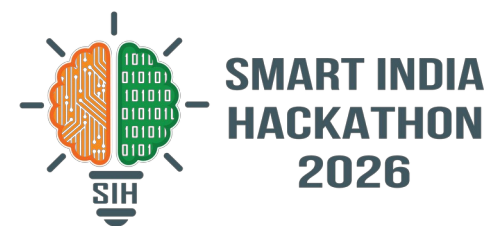


SMART INDIA HACKATHON 2026



FUND-TRAIL

Problem Statement ID – 26183

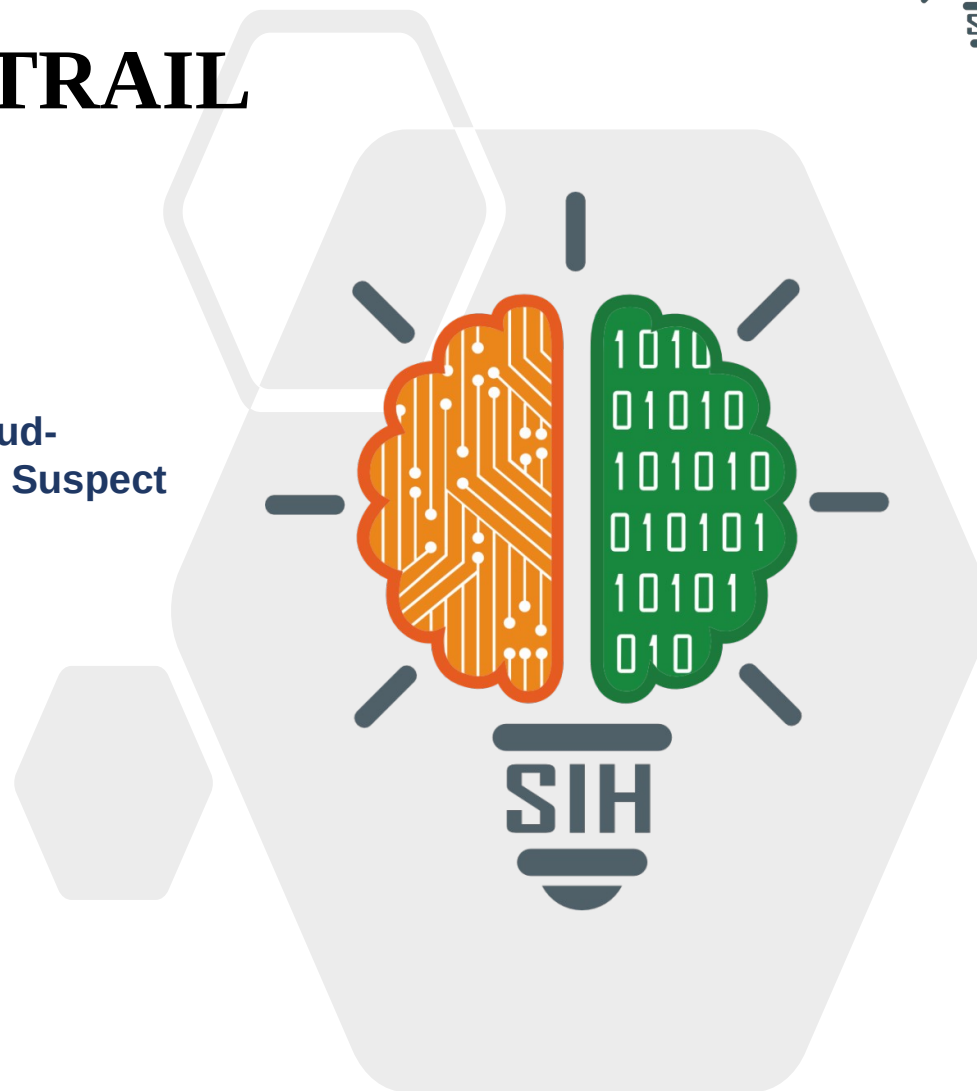
Problem Statement Title – Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addresses

Theme – Blockchain & Cybersecurity

PS Category – Software

Team ID –

Team Name – (Registered on portal)



PROPOSED SOLUTION

- One wallet address in — the likely cash-out exchange out
- Live chain data → fund-flow graph → actor clusters → confidence score
- One click produces a hash-sealed PDF for the case file

HOW IT ADDRESSES THE PROBLEM

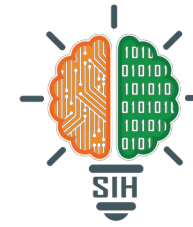
- Manual hop-by-hop tracing takes hours — scammers cash out in hours
- A wallet cannot be frozen; an exchange account can
- We name that exchange in minutes, not days

INNOVATION AND UNIQUENESS

- The missing middle: free and self-hostable, not enterprise-priced
- Explainable rules, not a black box — every point of the score is auditable
- Built around India's 1930 / cybercrime.gov.in complaint workflow

Your
Team
Name

TECHNICAL APPROACH



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TECHNOLOGIES USED

1. Python · FastAPI · NetworkX graph engine
2. Blockchain (BTC) · Etherscan (ETH) + local cache
3. React · Vite · Tailwind · force-directed graph
4. PostgreSQL / SQLite · SHA-256 evidence hash
5. GraphSense TagPacks · OFAC SDN list

No GPU. No model training — runs on one laptop.

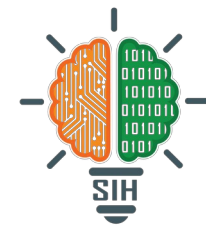
METHODOLOGY



IMPLEMENTATION STEPS

1. **CASE INTAKE** — wallet address, chain and complaint ID open a case
2. **GRAPH & CLUSTERING** — hundreds of addresses collapse into a few real actors
3. **DETECTION & SCORING** — five laundering patterns, each with a visible weight
4. **ATTRIBUTION & REPORT** — ranked exit exchange, human review, hash-sealed PDF

FEASIBILITY AND VIABILITY



ANALYSIS OF FEASIBILITY

- Bitcoin and Ethereum ledgers are public and free to query
- Graph work and rules, not deep learning — no GPU, no training
- Attribution data (TagPacks, OFAC SDN) is openly licensed

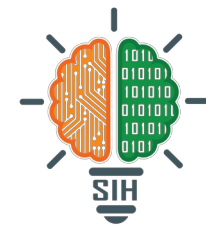
POTENTIAL CHALLENGES AND RISKS

- Free-tier API rate limits can break a live demo
- Mixers and cross-chain bridges end deterministic tracing
- Open tag lists miss smaller exchanges — some traces dead-end

STRATEGIES FOR OVERCOMING THESE CHALLENGES

- Cache-first design: pre-fetched demo wallets, rate limiter
- Report a bridge or mixer hop as a confidence boundary, never a guess
- Every attribution tag carries a public provenance link

IMPACT AND BENEFITS

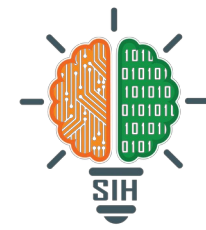


POTENTIAL IMPACT ON THE TARGET AUDIENCE

- Specialist tracing for cells that cannot afford it
- Hours of manual work reduced to minutes
- Better odds of freezing funds in time
- Court-usable evidence, not a screenshot

BENEFITS OF THE SOLUTION

- **Social** — recourse for ordinary scam victims
- **Economic** — free, self-hostable, no licence cost
- **Operational** — case workflow with human review
- **Legal** — explainable score, sealed evidence



DETAILS / LINKS OF THE REFERENCE AND RESEARCH WORK

- MHA — National Cybercrime Reporting Portal & 1930 helpline · cybercrime.gov.in
- Meiklejohn et al., “A Fistful of Bitcoins,” ACM IMC 2013 · [address-clustering heuristics](#)
- GraphSense TagPacks — open attribution tags · github.com/graphsense/graphsense-tagpacks
- OFAC SDN — sanctioned crypto addresses · ofac.treasury.gov
- Blockchair API (Bitcoin) · blockchair.com/api/docs
- Etherscan API V2 (Ethereum) · docs.etherscan.io
- GraphSense · BlockSci · [open-source blockchain forensics platforms](#)
- Chainalysis Crypto Crime Report · [laundering typologies](#)